VTPEH 6270 - Check Point 04

Data Simulation

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1 Introduction

This document is for instructions only. Please create your own R markdown document as a basis for the rest of the project. The document should respect the following criteria: • Is a PDF. • Include headers as relevant, and a table of content. • Include all R code. • Include all figures. • Console output may or may not be printed, as relevant. • Is understandable and pleasant to navigate (e.g., avoid printing full data set within the report) - Please note that formatting errors will be penalized Sections from previous reports can be re-used if relevant. Any section of the report can be revised as the project develops. The goal of this Check Point is to gain experience with data simulation in the context of power analyses.

2 Objective and Approach

2.1 Specific Objective

Is the rate of primary care physicians associated with the percentage of adults reporting fair or poor health at the county level in the United States?

2.2 Data Description

```r
library(readr)
library(dplyr)
library(tibble)
library(knitr)
library(tidyr)
library(ggplot2)
library(kableExtra)
library(purrr)
library(here)

df <- read_csv(here("data", "processed", "6270_Dataset_Selection_Luo_20260204.csv"))

infer_variable_type <- function(x) {
  x_no_na <- x[!is.na(x)]
  cls <- class(x)[1]

  # Date / POSIXt
  if (inherits(x, "Date") || inherits(x, "POSIXt")) return("date/time")

  # categorical
  if (is.factor(x) || is.character(x) || is.logical(x)) return("categorical")

  # numeric/integer -> continuous vs discrete
  if (is.integer(x)) return("discrete")

  if (is.numeric(x)) {
    n_unique <- length(unique(x_no_na))
    if (n_unique <= 10) return("discrete")
    return("continuous")
  }

  return("other")
}

data_description <- tibble(
  Variable_Name = names(df),
  Variable_Type_in_Dataset = sapply(df, infer_variable_type),
  R_Class = sapply(df, function(x) paste(class(x), collapse = ", "))
)
data_description_clean <- data_description %>%
  rename(
    "Variable Name" = Variable_Name,
    "Variable Type in Dataset" = Variable_Type_in_Dataset,
    "R Class" = R_Class
  )
kable(data_description_clean,
caption = "Description of Variables in the Dataset")
```

Table 1: Description of Variables in the Dataset

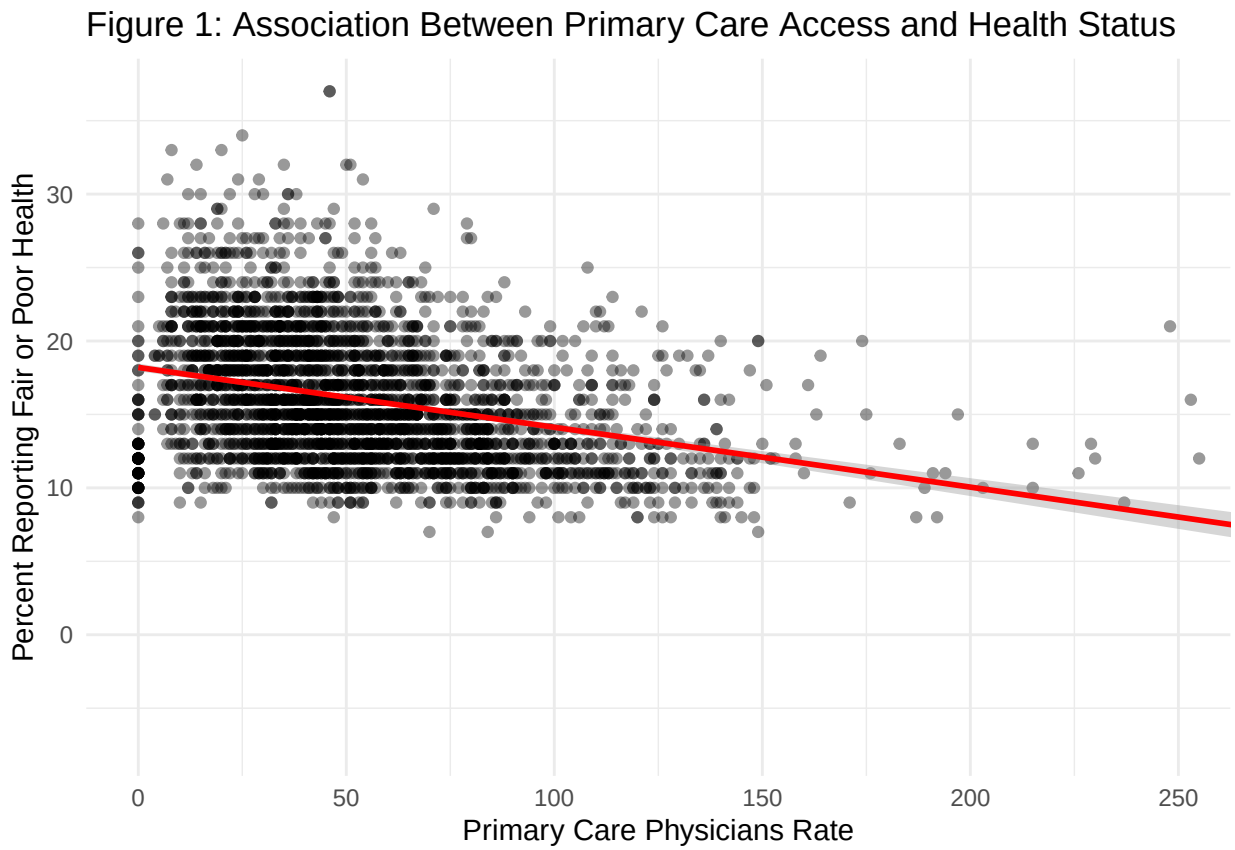
Variable Name	Variable Type in Dataset	R Class
FIPS	categorical	character
State	categorical	character
County	categorical	character
Population	continuous	numeric
% 65 and Over	continuous	numeric
% female	continuous	numeric
# Black	continuous	numeric
% Black	continuous	numeric
# American Indian & Alaska Native	continuous	numeric
% American Indian & Alaska Native	continuous	numeric
# Asian	continuous	numeric
% Asian	continuous	numeric
# Hispanic	continuous	numeric
% Hispanic	continuous	numeric
# Native Hawaiian/Other Pacific Islander	continuous	numeric
% Native Hawaiian/Other Pacific Islander	continuous	numeric
# Non-Hispanic white	continuous	numeric
% Non-Hispanic white	continuous	numeric
Median Household Income	continuous	numeric
# Unemployed	continuous	numeric
% Unemployed	continuous	numeric
# Some College	continuous	numeric
% Some College	continuous	numeric
# Primary Care Physicians	continuous	numeric
Primary Care Physicians Rate	continuous	numeric
% With Access to Exercise Opportunities	continuous	numeric
% Uninsured	continuous	numeric
Food Environment Index	continuous	numeric
% Adults Reporting Currently Smoking	continuous	numeric
% Adults with Obesity	continuous	numeric
% Fair or Poor Health	continuous	numeric

2.3 Data Visualization

```
df_clean <- df %>%
  select(`Primary Care Physicians Rate`,
         `% Fair or Poor Health`) %>%
  drop_na()

ggplot(df_clean,
       aes(x = `Primary Care Physicians Rate`,
           y = `% Fair or Poor Health`)) +
  geom_point(alpha = 0.4) +
  geom_smooth(method = "lm", color = "red", se = TRUE) +
  labs(
    x = "Primary Care Physicians Rate",
    y = "Percent Reporting Fair or Poor Health",
    title = "Figure 1: Association Between Primary Care Access and Health Status"
```

```
) + coord_cartesian(xlim = c(0, 250)) +
theme_minimal()
```



2.4 Description of Plausible Relationships

Visual inspection suggests a negative linear relationship between primary care physician rate and the percentage of adults reporting fair or poor health. Counties with higher physician availability tend to exhibit lower percentages of self-reported poor health. The relationship appears moderately strong, though substantial variability remains across counties.

2.5 Parameters of Interest

Table 2: Parameters of Interest for the Linear Model

Parameter	Symbol	Estimate	Description
Slope	a	-0.041	Change in percent reporting fair/poor health per one-unit increase in primary care physician rate
Intercept	b	18.202	Expected percent reporting fair/poor health when physician rate equals zero
Noise (Residual SD)	ε	4.151	Residual variability in percent reporting fair/poor health across counties not explained by physician rate

Based on the empirical data, the estimated slope is approximately -0.04, indicating that higher physician availability is associated with lower prevalence of fair or poor health. The residual standard deviation provides an estimate of unexplained variability and will inform the noise parameter in the simulation.

3 Simulation

3.1 Simulation Basis

```r
set.seed(123)

# Baseline parameters estimated from the empirical data
a <- -0.040747
b <- 18.201724
sigma <- 4.151
n <- 3000

# Simulate X using the empirical distribution
x_sim <- sample(df_clean$`Primary Care Physicians Rate`, size = n, replace = TRUE)

# Simulate noise
eps <- rnorm(n, mean = 0, sd = sigma)

# Generate Y from the linear relationship
y_sim <- b + a * x_sim + eps

# Create simulated dataset
sim_data <- data.frame(
  `Primary Care Physicians Rate` = x_sim,
  `% Fair or Poor Health` = y_sim,
  check.names = FALSE
)
# Preview
head(sim_data)
```

3.2 Simulation Output

```r
sim_summary <- sim_data %>%
  mutate(`PCP Rate Quartile` = ntile(`Primary Care Physicians Rate`, 4)) %>%
  group_by(`PCP Rate Quartile`) %>%
  summarise(
    `Mean (%)` = mean(`% Fair or Poor Health`),
    `SD (%)` = sd(`% Fair or Poor Health`),
    N = n()
  )

knitr::kable(sim_summary,
             caption = "Simulated Summary Statistics by PCP Rate Quartile")
```

Table 3: Simulated Summary Statistics by PCP Rate Quartile

PCP Rate Quartile	Mean (%)	SD (%)	N
1	17.46705	4.068589	750
2	16.54537	4.171963	750
3	15.69762	4.059634	750
4	13.94549	4.409507	750

3.3 Simulation Function

```
simulate_pcp_health <- function(effect_size, noise, sample_size, seed = NULL) {

  if (!is.null(seed)) set.seed(seed)
  x_sim <- sample(df_clean$`Primary Care Physicians Rate`,
                  size = sample_size,
                  replace = TRUE)
  b <- 18.201724
  eps <- rnorm(sample_size, mean = 0, sd = noise)
  y_sim <- b + effect_size * x_sim + eps
  y_sim <- pmin(pmax(y_sim, 0), 100)

  #Simulated dataset
  sim_data <- data.frame(
    `Primary Care Physicians Rate` = x_sim,
    `% Fair or Poor Health` = y_sim,
    check.names = FALSE
  )


  sim_summary <- sim_data %>%
    mutate(quartile = ntile(`Primary Care Physicians Rate`, 4)) %>%
    group_by(quartile) %>%
    summarise(
      `Mean (%)` = mean(`% Fair or Poor Health`),
      `SD (%)` = sd(`% Fair or Poor Health`),
      N = n(),
      .groups = "drop"
    ) %>%
    rename(`PCP Rate Quartile` = quartile)

  return(list(data = sim_data, summary = sim_summary))
}
```

3.4 Simulation Automation

```
# Parameter grids
effect_sizes <- seq(-0.08, -0.01, length.out = 10)
sample_sizes <- round(seq(300, 4500, length.out = 10))
noise_levels <- c(3, 4.151, 8)
```

```r
length(effect_sizes); length(sample_sizes); length(noise_levels)

# Storage list
sim_results <- vector("list", length(effect_sizes) * length(sample_sizes) * length(noise_levels))
k <- 1

# Run all simulations
for (a in effect_sizes) {
  for (n in sample_sizes) {
    for (sig in noise_levels) {

      run <- simulate_pcp_health(
        effect_size = a,
        noise = sig,
        sample_size = n
      )

      # store everything (data + summary + parameters)
      sim_results[[k]] <- list(
        effect_size = a,
        sample_size = n,
        noise = sig,
        data = run$data,
        summary = run$summary
      )

      k <- k + 1
    }
  }
}

# Name each run
names(sim_results) <- sapply(sim_results, function(x) {
  paste0("a=", round(x$effect_size, 3),
         "_n=", x$sample_size,
         "_sigma=", round(x$noise, 3))
})

# Quick sanity check
length(sim_results)
names(sim_results)[1:5]
sim_results[[1]]$summary
```

4 Visualization

```r
plot_df <- map_dfr(sim_results, function(run) {

  s <- run$summary
  # quartile 1 and 4 mean
  mean_q1 <- s$`Mean (%)`[s$`PCP Rate Quartile` == 1]
```

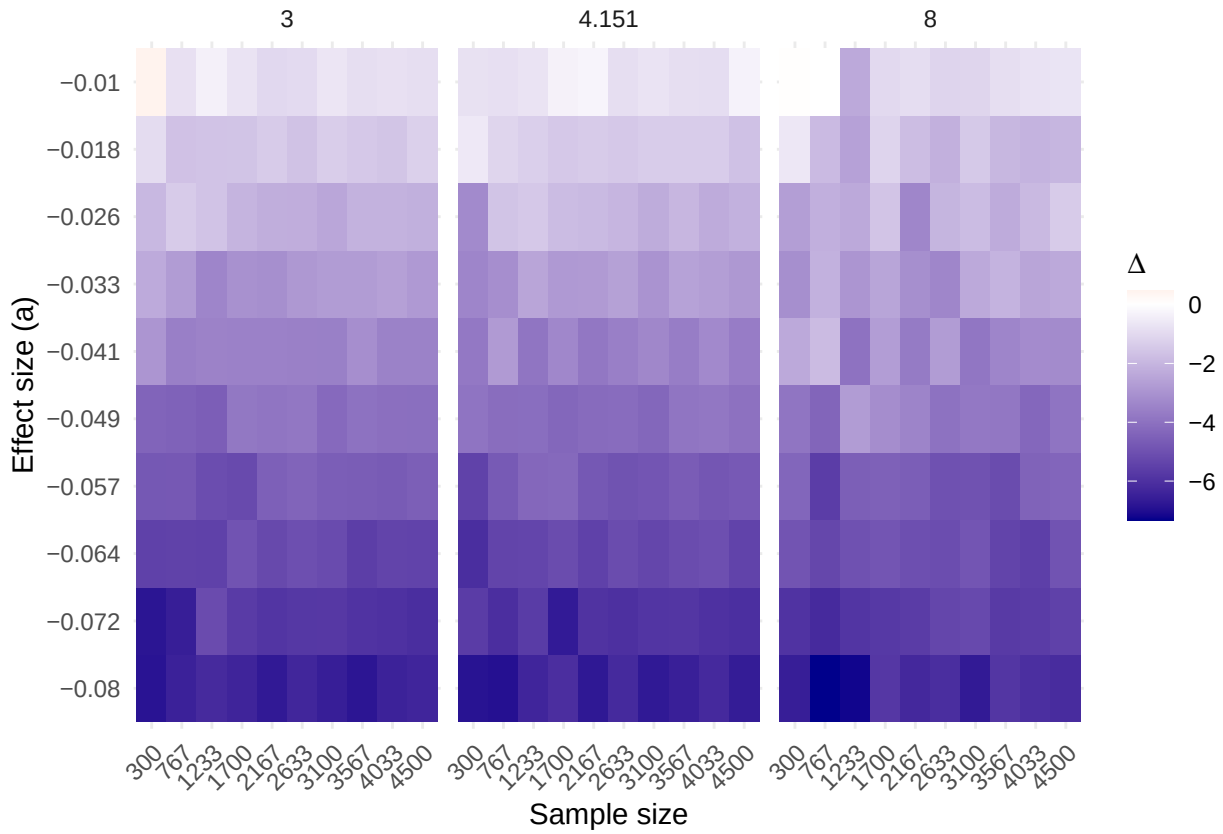
```r
  mean_q4 <- s$`Mean (%)`[s$`PCP Rate Quartile` == 4]

  tibble(
    effect_size = run$effect_size,
    sample_size = run$sample_size,
    noise = run$noise,
    Delta = mean_q4 - mean_q1
  )
})

head(plot_df)
```

```
## # A tibble: 6 x 4
##   effect_size sample_size noise Delta
##         <dbl>       <dbl> <dbl> <dbl>
## 1       -0.08         300  3    -6.87
## 2       -0.08         300  4.15 -6.89
## 3       -0.08         300  8    -6.57
## 4       -0.08         767  3    -6.48
## 5       -0.08         767  4.15 -6.95
## 6       -0.08         767  8    -7.34
```

```r
ggplot(plot_df, aes(x = factor(sample_size),
                    y = factor(effect_size),
                    fill = Delta) )+
  geom_tile() +
  facet_wrap(~ noise) +
  scale_fill_gradient2(low = "darkblue",
                       mid = "white",
                       high = "red",
                       midpoint = 0) +
  theme_minimal() + theme(
  axis.text.x = element_text(angle = 45, hjust = 1)
) +
  labs(
    x = "Sample size",
    y = "Effect size (a)",
    fill = expression(Delta)
  ) + scale_y_discrete(labels = function(x) round(as.numeric(x), 3))
```



5 Interpretation

The heatmap summarizes how the estimated difference between the highest and lowest PCP rate quartiles (Δ = Mean(Y)Q4 - Mean(Y)Q1) varies across effect sizes, sample sizes, and noise levels. As the effect size becomes more negative, the magnitude of Δ increases consistently, indicating a stronger difference in the outcome between the highest and lowest PCP rate quartiles. When the effect size is close to zero, Δ remains small across all parameter combinations, suggesting little difference in outcome across PCP rate groups.

Sample size mainly influences the stability of the estimates. Larger sample sizes produce smoother and more consistent patterns, while smaller sample sizes show slightly greater variability. Increasing noise also introduces more variability in the estimated differences, though the overall trend remains visible. Overall, the simulation indicates that effect size drives the strength of the observed relationship, while sample size and noise primarily affect estimation variability.

6 AI Use Disclosure Statement

AI assistance (ChatGPT) was used to support understanding of statistical concepts, interpretation of results, and improvement of writing clarity. All analyses, interpretations, and final responses were reviewed and verified by the student.